

ANTICHAIN SUBSET SUMS IN RANK-THREE MULTIPLICATIVE SEMIGROUPS

ABSTRACT. Let $a, b, c \geq 2$ be pairwise coprime and $\mathcal{S} = \{a^k b^\ell c^m : k, \ell, m \geq 0\}$. For every real $1 < \rho < \min(a, b, c)$, every sufficiently large integer is a subset sum of $\mathcal{S} \cap [x, \rho x)$ for some x . Such a set is automatically a divisibility antichain, resolving Erdős Problem 123. For rational ρ we prove the stronger statement that the Bernoulli subset sums of $\mathcal{S} \cap [x, \rho x)$ satisfy a uniform local central limit theorem. The main new ingredient is an arithmetic rigidity argument on a triangular grid in exponent space; the remaining Fourier analysis is standard.

1. STATEMENT AND REDUCTION

Fix pairwise-coprime integers $a, b, c \geq 2$, and write

$$\mathcal{S} = \{a^k b^\ell c^m : k, \ell, m \in \mathbb{Z}_{\geq 0}\}, \quad d = \min(a, b, c).$$

Let $\rho = p/q \in \mathbb{Q}$ satisfy $1 < \rho < d$, with $p, q \in \mathbb{N}$ and $q > 0$. For an integer $x \geq 1$ set

$$\mathcal{S}_x = \mathcal{S} \cap [x, \rho x), \quad W_x = \sum_{s \in \mathcal{S}_x} s, \quad V_x^2 = \sum_{s \in \mathcal{S}_x} s^2,$$

and

$$\mu_x = \frac{W_x}{2}, \quad \sigma_x = \frac{V_x}{2}.$$

Let $(\xi_s)_{s \in \mathcal{S}_x}$ be independent Bernoulli variables taking the values 0, 1 with equal probability, and put $Y_x = \sum_{s \in \mathcal{S}_x} s \xi_s$.

Theorem 1.1 (Local limit theorem). *As $x \rightarrow \infty$ through the positive integers,*

$$\mathbb{P}(Y_x = n) = \frac{1}{\sqrt{2\pi} \sigma_x} \exp\left(-\frac{(n - \mu_x)^2}{2\sigma_x^2}\right) + o\left(\frac{1}{\sigma_x}\right), \quad (1)$$

uniformly for $n \in \mathbb{Z}_{\geq 0}$. In particular, for all sufficiently large x , every integer n with $|n - \mu_x| \leq \sigma_x$ is a subset sum of $\mathcal{S}_x$.

Theorem 1.2 (Erdős Problem 123). *For every real $1 < \rho < d$, every sufficiently large integer N can be written as*

$$N = \sum_{s \in A} s, \quad A \subset \mathcal{S} \cap [x, \rho x)$$

for some positive integer x . Consequently no two distinct elements of A divide one another.

The last assertion is immediate. If $u < v$ belong to $\mathcal{S}$ and $u \mid v$, unique factorization in the three pairwise-coprime bases gives $v/u = a^r b^s c^t \geq d$; two elements of $[x, \rho x)$ have ratio less than $\rho < d$.

We prove Theorem 1.1 for rational ρ , then obtain the theorem for an arbitrary real ρ by choosing a rational ρ' with $1 < \rho' < \rho$.

2. A TRIANGULAR GRID IN THE EXPONENT SLAB

Put

$$\alpha = \log a, \quad \beta = \log b, \quad \gamma = \log c, \quad \eta = \log \rho,$$

and, with $L = \log x$,

$$\mathcal{E}_L = \{(k, \ell, m) \in \mathbb{Z}_{\geq 0}^3 : L \leq \alpha k + \beta \ell + \gamma m < L + \eta\}.$$

This set parametrizes $\mathcal{S}_x$. The ratios of any two of α, β, γ are irrational: for instance $\alpha/\beta = r/s \in \mathbb{Q}$ would give $a^s = b^r$, contrary to $\gcd(a, b) = 1$.

We use the following elementary consequence of irrational rotation. If $u, v > 0$, $u/v \notin \mathbb{Q}$, and $0 < \eta < v$, then there is R such that, whenever

$$uX + vY = A + \frac{\eta}{2}, \quad X, Y \geq R,$$

there are nonnegative integers r, s , each within R of the corresponding real coordinate, for which

$$A \leq ur + vs < A + \eta. \quad (2)$$

Indeed, bounded multiples of u/v form a finite $\eta/(8v)$ -net modulo 1. Writing $r_0 = \lfloor X \rfloor$ and $y = (A - ur_0)/v$, choose a bounded integer h so that the fractional part of $y - hu/v$ lies between $1 - 5\eta/(8v)$ and $1 - 3\eta/(8v)$; then take $r = r_0 + h$ and $s = \lceil y - hu/v \rceil$. This gives (2), and the displayed linear relation bounds the error in the second coordinate.

Applying this to the three pairs of logarithms gives a constant R_0 with the following property: every point (X, Y, Z) on

$$\alpha X + \beta Y + \gamma Z = L + \frac{\eta}{2} \quad (3)$$

whose positive coordinates are at least $2R_0$, and of which at most one is zero, lies within R_0 coordinatewise of a point of $\mathcal{E}_L$; a zero coordinate remains zero. When all coordinates are positive, round Y to a nearest integer ℓ_0 , replace Z by $Z + (\beta/\gamma)(Y - \ell_0)$, and apply (2) to α, γ . If one coordinate is zero, apply the two-variable statement directly to the other two.

Proposition 2.1 (Geometry and moments). *There are constants $c, C > 0$ and integers $D, K \geq 1$ such that, for all large x ,*

$$cL^2 \leq |\mathcal{S}_x| \leq CL^2, \quad cxL \leq V_x \leq CxL, \quad cxL^2 \leq W_x \leq CxL^2. \quad (4)$$

Moreover, for some $n \asymp L$ there are distinct elements $s_v \in \mathcal{S}_x$ indexed by

$$\Delta_n^\circ = \{(i, j, r) \in \mathbb{Z}_{\geq 0}^3 : i + j + r = n\} \setminus \{(n, 0, 0), (0, n, 0), (0, 0, n)\},$$

such that a zero coordinate of v gives a zero corresponding exponent of s_v , and adjacent vertices satisfy

$$A_{vw}s_v = B_{vw}s_w, \quad 1 \leq A_{vw}, B_{vw} \leq K. \quad (5)$$

Proof. Choose τ_0 so large that $\tau_0/\max(\alpha, \beta, \gamma) > 4R_0$. For large L , let

$$n = \left\lfloor \frac{L + \eta/2}{\tau_0} \right\rfloor, \quad \tau = \frac{L + \eta/2}{n};$$

then $n \asymp L$ and $\tau_0 \leq \tau \leq 2\tau_0$. For $v = (i, j, r) \in \Delta_n^\circ$, round the point

$$\left(\frac{\tau i}{\alpha}, \frac{\tau j}{\beta}, \frac{\tau r}{\gamma} \right)$$

to an exponent triple in $\mathcal{E}_L$. The corners were removed, so the rounding lemma applies. It preserves the coordinate faces. Distinct vertices have one target coordinate separated by more than $4R_0$, hence have distinct images. Adjacent targets differ by a bounded amount, so their rounded exponents differ coordinatewise by at most some fixed D . Clearing the positive and negative exponent differences gives (5) with $K = (abc)^D$.

Since $|\Delta_n^\circ| \asymp n^2$, this gives $|\mathcal{S}_x| \gg L^2$. Conversely, $k, \ell \ll L$, while for fixed k, ℓ there is at most one m , because two choices would change the logarithmic weight by at least $\gamma > \eta$. Thus $|\mathcal{S}_x| \ll L^2$. Finally $x \leq s < \rho x$ for every $s \in \mathcal{S}_x$, and (4) follows by summing s and s^2 . $\square$

3. LOW-ENERGY FREQUENCIES

For $t \in \mathbb{T} = \mathbb{R}/\mathbb{Z}$, put

$$Q_x(t) = \sum_{s \in \mathcal{S}_x} \|st\|^2, \quad \|y\| = \min_{m \in \mathbb{Z}} |y - m|.$$

The following is the arithmetic core of the proof.

Proposition 3.1 (Rigidity). *There are constants $z_0, C, \kappa, \delta > 0$ such that, for all large x and $0 \leq z \leq z_0 L^2$,*

$$\text{meas}\{t \in \mathbb{T} : Q_x(t) \leq z\} \leq \frac{1}{x} \exp\left(C \left(1 + \frac{z}{L}\right) \log(L + 2)\right). \quad (6)$$

The constants may be chosen so that $\delta d \leq 1/8$ and

$$Q_x(t) < \kappa L \implies \|t\| \leq \frac{\delta}{x}. \quad (7)$$

Proof. Use the grid from Proposition 2.1, and choose $0 < \delta_0 < \min(1/(4K), 1/32)$. For each vertex let m_v be a nearest integer to $s_v t$, and call v bad if $|s_v t - m_v| > \delta_0$. If there are H bad vertices, then

$$H\delta_0^2 \leq Q_x(t). \quad (8)$$

For $n/3 \leq q \leq 2n/3$, consider the row

$$R_q = \{(i, q - i, n - q) : 0 \leq i \leq q\}.$$

There are $\gg n$ disjoint such rows, so one contains at most $4H/n$ bad vertices. From its central half consider the disjoint paths

$$P_j = \{(q - j + r, j, n - q - r) : 0 \leq r \leq n - q\}, \quad q/4 \leq j \leq 3q/4.$$

One of them contains at most $7H/n$ bad vertices. Their union is connected, has three endpoints on the three coordinate faces, and contains only $O(1 + H/n)$ bad vertices. If $8H < n$, the row and path can both be chosen with no bad vertices.

Orient the edges of this row and path. For an edge vw , define the integer

$$e_{vw} = B_{vw}m_w - A_{vw}m_v,$$

where (5) is used. If both endpoints are good, then

$$|e_{vw}| \leq B_{vw}|m_w - s_w t| + A_{vw}|m_v - s_v t| < 2K\delta_0 < 1,$$

so $e_{vw} = 0$. Thus, when $Q_x(t) \leq z$, only $O(1 + z/L)$ edge values can be nonzero, by (8) and $n \asymp L$. The row, path, positions of the nonzero edge values, and their bounded integer values can therefore take at most

$$\exp\left(C\left(1 + \frac{z}{L}\right)\log(L + 2)\right) \quad (9)$$

possible values.

Fix these data and compare two compatible frequencies. If Δ_v denotes the difference of their nearest integers, then

$$B_{vw}\Delta_w = A_{vw}\Delta_v, \quad B_{vw}s_w = A_{vw}s_v.$$

Connectivity gives $\Delta_v = rs_v$ for one rational number r . The three end weights respectively omit a, b, c ; pairwise coprimality therefore gives their gcd, and hence the gcd of all weights on the row and path, as 1. Consequently $r \in \mathbb{Z}$. At a fixed root of weight s_0 , the nearest integer lies between 0 and s_0 , so a fixed choice of the edge data permits at most three root values modulo s_0 . Each root value confines t to an interval of length $1/s_0 \leq 1/x$. Multiplying this by (9) proves (6).

If $Q_x(t) < \kappa L$ and κ is sufficiently small, then (8) gives $8H < n$. Choose a completely good row and path. Every e_{vw} is zero, so m_v/s_v is constant. Its denominator divides the gcd of the three face weights, which is 1; hence the common ratio is an integer. At any vertex, $|s_v t - m_v| \leq \delta_0$ and $s_v \geq x$, so initially $\|t\| \leq \delta_0/x$. To sharpen the constant, choose a rational ρ_0 with $1 < \rho_0 \leq \min(\rho, 3/2)$ and repeat the preceding argument for $\mathcal{S} \cap [x, \rho_0 x]$. Writing $t = r + u$, we then have $|u| \leq 1/(32x)$; nearest-integer rounding vanishes on this set, and hence $Q_x(t) \geq u^2 \sum s^2 \gg x^2 L^2 u^2$. Thus $Q_x(t) < \kappa L$ forces $|u| \leq \delta/x$ for any fixed $\delta > 0$ once x is large. Choose $\delta d \leq 1/8$ to obtain (7). $\square$

4. FOURIER ANALYSIS AND COMPLETION OF THE PROOF

We shall use

$$|\cos(\pi y)| \leq \exp(-2\|y\|^2), \quad (10)$$

which follows from $\sin(\pi u/2) \geq u$ for $0 \leq u \leq 1/2$.

Lemma 4.1 (Fourier tails).

$$\int_{T/V_x \leq |t| \leq 1/2} \left| \prod_{s \in \mathcal{S}_x} \cos(\pi s t) \right| dt = o\left(\frac{1}{V_x}\right), \quad T = L^{1/4}. \quad (11)$$

Proof. For large x , $T/V_x < 2\delta/x$. On $T/V_x \leq |t| \leq 2\delta/x$, every $s \in \mathcal{S}_x$ satisfies $|st| < 2\delta d \leq 1/4$, so (10) and $V_x^2 = \sum s^2$ give

$$\prod_{s \in \mathcal{S}_x} |\cos(\pi s t)| \leq \exp(-2V_x^2 t^2).$$

Its integral is at most $V_x^{-1} \int_T^\infty e^{-2u^2} du = o(1/V_x)$.

On $2\delta/x \leq |t| \leq 1/2$, Proposition 3.1 gives $Q_x(t) \geq \kappa L$. By (10), the integrand is at most $e^{-2Q_x(t)}$. Layering by the values of Q_x and using (6) yields

$$\int e^{-2Q_x(t)} dt \leq \frac{2}{x} \int_{\kappa L}^{z_0 L^2} e^{-2z + C(1+z/L) \log(L+2)} dz + e^{-2z_0 L^2}.$$

For large L this is $O(x^{-1} L^C e^{-3\kappa L/2}) + e^{-2z_0 L^2}$, which is $o(1/(xL)) = o(1/V_x)$ by (4). $\square$

Proof of Theorem 1.1. Fourier inversion gives

$$\mathbb{P}(Y_x = n) = \int_{-1/2}^{1/2} e^{-2\pi i(n-\mu_x)t} \prod_{s \in \mathcal{S}_x} \cos(\pi st) dt. \quad (12)$$

On $|t| \leq T/V_x$, all $|\pi st|$ tend uniformly to zero. The elementary estimates

$$|\cos u - e^{-u^2/2}| \leq u^4, \quad \left| \prod_j x_j - \prod_j y_j \right| \leq \sum_j |x_j - y_j| \quad (|x_j|, |y_j| \leq 1)$$

give

$$\left| \prod_{s \in \mathcal{S}_x} \cos(\pi st) - \exp\left(-\frac{\pi^2 V_x^2 t^2}{2}\right) \right| \leq \pi^4 t^4 \sum_{s \in \mathcal{S}_x} s^4. \quad (13)$$

Since $s < px$ and $\sum s^2 = V_x^2$, $\sum s^4 \leq (px)^2 V_x^2$. Integrating the right side of (13) over $|t| \leq T/V_x$ gives

$$O\left(\frac{(px)^2 T^5}{V_x^3}\right) = o\left(\frac{1}{V_x}\right)$$

by (4). The Gaussian tail outside this interval is also $o(1/V_x)$. Hence the principal part of (12) equals

$$\int_{\mathbb{R}} e^{-2\pi i(n-\mu_x)t} \exp\left(-\frac{\pi^2 V_x^2 t^2}{2}\right) dt + o\left(\frac{1}{V_x}\right).$$

The standard Gaussian Fourier transform evaluates this integral as

$$\frac{\sqrt{2/\pi}}{V_x} \exp\left(-\frac{2(n-\mu_x)^2}{V_x^2}\right) = \frac{1}{\sqrt{2\pi} \sigma_x} \exp\left(-\frac{(n-\mu_x)^2}{2\sigma_x^2}\right).$$

The error is uniform in n , since the omitted oscillatory factor always has modulus 1. Lemma 4.1 proves (1).

If $|n - \mu_x| \leq \sigma_x$, the main term in (1) is at least $e^{-1/2}/(\sqrt{2\pi}\sigma_x)$, whereas the error is $o(1/\sigma_x)$. Thus $\mathbb{P}(Y_x = n) > 0$ for all sufficiently large x , so n is a subset sum of $\mathcal{S}_x$. $\square$

Proof of Theorem 1.2. First suppose $\rho = p/q$ is rational. By Proposition 2.1, $|\mathcal{S}_x| \geq 4p^4$ for all sufficiently large x , and hence

$$V_x^2 \geq 4p^4 x^2, \quad V_x \geq 2p^2 x. \quad (14)$$

Also

$$W_{x+1} \leq W_x + p^2(x+1). \quad (15)$$

Indeed, an element entering when x is replaced by $x+1$ lies in an interval of length $p/q < p$, so there are at most p such integers, each at most $p(x+1)$; elements leaving $\mathcal{S}_x$ only decrease the sum.

Fix a sufficiently large initial X . For N large enough that $W_X \leq 2N$, choose the largest $x \geq X$ with $W_x \leq 2N$. Such an x exists and is finite, since $\mathcal{S}_x$ is nonempty for all sufficiently large x , and then $W_x \geq x$. By maximality and (15),

$$0 \leq 2N - W_x < W_{x+1} - W_x \leq p^2(x+1) \leq 2p^2 x \leq V_x.$$

Thus $|N - \mu_x| \leq \sigma_x$, and Theorem 1.1 gives a subset of $\mathcal{S}_x$ summing to N .

For a real $1 < \rho < d$, choose a rational ρ' with $1 < \rho' < \rho$. The rational case supplies the representation inside $[x, \rho'x) \subset [x, \rho x)$. As observed at the start, every such subset is a divisibility antichain. $\square$